

Kaan Baha Sever

SOFTWARE DEVELOPER | MATH-DRIVEN SOLUTIONS & ALGORITHMS

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Software developer with a mathematics background (Istanbul University). I build systems software in modern C++ and Go: desktop agents, telemetry and messaging platforms, and algorithm-driven applications, together with the CI/CD pipelines that test, package and ship them. From 2021 to 2024 I owned the development lifecycle of a web platform at crowd.inc, from database schema to Linux servers.

EXPERIENCE

Software Developer · crowd.inc Jul 2021 – Mar 2024

Owned the software development lifecycle end to end, from relational schema design in PostgreSQL to RESTful APIs in Python and Flask with a jQuery front end.

- Designed granular role-based access control (RBAC) with strict permission boundaries between public and private data.
- Built scalable feed pagination and dynamic data loading across the API and the front end.
- Provisioned and configured the Linux servers the platform ran on.
- Kept the system reliable through rigorous unit and integration testing.

Python · Flask · PostgreSQL · jQuery · Linux

SELECTED PROJECTS

Asion · asion.app Since 2024 · in development, early access

A privacy-first, cross-platform productivity and workstation activity tracker for individuals and teams, and for academic time tracking, built with self-hosting in mind. It runs on native OS event hooks on macOS, Linux and Windows, with a lightweight daemon architecture, gRPC/Protobuf IPC and an encrypted SQLCipher store. I built its multi-platform GitHub Actions pipelines, with cross-compilation runners, for one-click testing, packaging, artefact generation and deployment.

C/C++ · Objective-C · Go · gRPC

Novacast · novacast.app Since 2025 · in production

Real-time message broadcasting and device orchestration built around low-latency pub/sub pipelines. Built to replace inadequate legacy software at a computer firm where I worked: prototyped in Python, then re-architected in late 2025 as a Go and MQTT microservice.

Go · MQTT

Karecik · github.com/KaanBahaSever/karecik Since 2021 · in production · GPL-3.0

A multi-tenant QR-menu SaaS that local cafés and restaurants use to manage menus and orders and to interact with customers, built for zero downtime and low latency. Early prototypes date from 2021 under earlier working titles; it shipped to production in 2026.

Go · PostgreSQL

Açık Matematik · acik-matematik.com Since 2023 · in production · open source

An academic publishing platform and modern textbook initiative for undergraduate mathematics in Turkish, with reproducible numerical computing. Conceived in 2023 and launched in June 2026; the open-source release was announced in September 2026.

Quarto · Markdown · Python

Rocket-Up · github.com/KaanBahaSever/Rocket-Up Since 2020 · in development · MIT

High-power rocket flight simulation: a narrow, purpose-built 2020 Python prototype, now being rewritten as a modular, open-source aerodynamic simulation engine in modern C++.

C++

NeoSMBIOS · github.com/KaanBahaSever/NeoSMBIOS Open source · MIT

A header-only, zero-copy, bounds-checked SMBIOS/DMI firmware-table parser for C++23.

C++

i18n-cpp · github.com/KaanBahaSever/i18n-cpp Open source · MIT

A lightweight, header-only C++ internationalisation library: translations kept in language files and looked up by key, with dynamic string interpolation.

C++

A bilingual (English/Turkish) static site with privacy-first browser tools that run entirely on the device (PDF, images, prime factorisation, great-circle distance), algorithmic games (a Battleship AI with hunt-and-target and probability-density modes, minimax tic-tac-toe) and technical writing.

TypeScript · Astro

Early work · github.com/KaanBahaSever/BattleShips

2016 – 2019

Learned programming in C# at a vocational high school from 2016, moving on to data structures and algorithms; by 2019 built database applications on Microsoft SQL Server, and desktop tic-tac-toe and Battleship games as graduation projects.

C# · SQL

EDUCATION

Bachelor of Science in Mathematics · Istanbul University · Faculty of Science

Nov 2019 – Present

- Preparing to graduate, with a grounding in pure mathematics: real analysis, abstract algebra and topology.
- Planning a Master's degree in Computer Science next.

LEADERSHIP & VOLUNTEERING

Vice President, Istanbul University Rocket Club

2019 – 2022

- With the club's team, designed and built one low-altitude rocket (5,000 ft) and two high-altitude rockets (10,000 ft).
- Solely developed the flight avionics firmware and the parachute deployment control system, both driven by onboard sensor fusion of orientation, gyroscope and altimeter data.
- Designed the SD-card telemetry logging protocol that worked alongside the RF telemetry modules, and built a desktop dashboard that parses flight data and plots post-flight trajectories.

Core Team Member, Google Developer Student Clubs

2023

- Hosted technical workshops and live streams on Flask, HTML and Git/GitHub.
- Organised Cyber Security Week, including a live-streamed technical interview with a CCIE-certified network security engineer.

Mathematics Club

Later university years

- Organised and ran academic events, including seminars and logic and mathematics competitions.

RESEARCH

Autonomous parachute guidance · github.com/KaanBahaSever/AutonomousParachute

A trajectory-steering algorithm in Python that uses linear algebra and atmospheric descent dynamics to guide a payload to a designated landing coordinate.

High school research project · TÜBİTAK (Scientific and Technological Research Council of Türkiye)

2019

- Researched cryptography and visual programming logic, including an early prototype of a flowchart-based visual coding tool.

SKILLS

Programming

C++ (C++23) · Go · C · Python · C# · Objective-C · TypeScript / JavaScript · SQL

Systems & networking

gRPC · Protobuf · MQTT · OS event hooks · Linux servers

DevOps & automation

GitHub Actions · Multi-platform CI/CD · Cross-compilation runners · One-click test, package & deploy · Bash · Batch · Python scripting

Databases

PostgreSQL · SQLite / SQLCipher · MySQL · MS SQL

Web

React · Astro · HTML / CSS · Flask · jQuery · REST APIs

Scientific computing

MATLAB · Mathematica · R

Version control

Git · GitHub

CERTIFICATES & ACTIVITIES

21st Century Competencies · YetGen · Mehmet Zorlu Foundation

2020

Erasmus+ Youth Exchange · Arrecife, Spain

Jun 2017 – Dec 2018

LANGUAGES

Turkish (native) · English